

Talha Omer

Senior Lecturer in Statistics

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Research Profile

Senior Lecturer in Statistics at Linnaeus University with a PhD in Statistics (Jönköping University). Research focuses on econometrics and statistical learning for mixed-frequency and high-dimensional models, shrinkage/regularization methods, Biostatistics and Statistical Methods for Health and Disease Modeling, and macro-financial forecasting, with applications including volatility forecasting and health-related count-data modeling.

Education

- **PhD in Statistics**, Jönköping University, Sweden 2019–2024
 Dissertation: *Shrinkage Estimation Methods for Mixed Data Sampling Regression and Heterogeneous Autoregressive Models*
 Department of Economics, Finance and Statistics, JIBS, Sweden
 Supervisors: Pär Sjölander; Kristofer Månsson; B. M. G. Golam Kibria; Daniel J. Henderson
 Defense: September 13, 2024
- **M.Phil. in Statistics**, Quaid-i-Azam University, Islamabad, Pakistan 2013–2015
 Thesis: *Optimization of Meteorological Monitoring Network Pakistan*
- **M.Sc. in Statistics**, PMAS University of Arid Agriculture, Rawalpindi, Pakistan 2011–2013
- **BA (Stat/Econ)**, Bahauddin Zakariya University, Multan, Pakistan 2009–2011
- **Diploma of Associate engineering (Eng/Phy/Math)**, PBTE, Lahore, Pakistan 2005–2008
- **Grade 10 (Phy/Chem/Elec)**, BISE, DG KHAN, Pakistan 2003–2005

Research Experience

1. **Research Fellow in Statistics**, Department of Economics, Finance and Statistics, Jönköping University, Sweden Sep 2024–May 2025
 Conduct research within the project “*Improving the Swedish Regulatory Model for Electricity Distribution*”. Contribute to the development and evaluation of econometric and statistical models for electricity distribution regulation, including efficiency analysis, cost modeling, and integration of advanced statistical and machine learning methods for regulatory assessment.
2. **Research Assistant**, HEC National Research Project University (NRPU), Pakistan Sep 2014– Sep 2015
 Worked on the project “*Optimization of Meteorological Monitoring Network in Pakistan*”. Responsible for statistical modeling, spatial analysis, and optimization of environmental monitoring station locations. The project was funded by the Higher Education Commission (HEC), Pakistan.

Academic Appointments

1. **Senior Lecturer in Statistics**, Department of Economics and Statistics, Linnaeus University, Växjö, Sweden 2025–Present
Master level courses: Statistical Machine Learning; Basic Statistics; Statistical Data Processing; Advanced Statistical Data Analysis.

Undergraduate level courses: Statistical Methods for Business and Economics; Statistics for Marketing Decisions; International Market Analysis and Statistics.

2. **Teaching Assistant**, Jönköping International Business School (JIBS), Jönköping University, Sweden 2019–2024

Master level courses: Econometrics

Undergraduate level courses: Econometrics I; Econometrics II; Business Statistics I; Business Statistics II; Introduction to Statistical Methods.

3. **Lecturer**, Department of Statistics and Computer Science, University of Veterinary and Animal Sciences, Lahore, Pakistan 2016–2019

Levels: Undergraduate, Master, and PhD

PhD course: Experimental Design.

Master level courses: Biostatistics; Applied Biostatistics; Experimental Statistics

Undergraduate level courses: Business Statistics; Statistics and Probability.

4. **Adjunct Lecturer**, Department of Mathematics and Statistics, Arid Agriculture University, Rawalpindi, Pakistan 2015–2016

Master level Courses: Computational Statistics; Statistics for Social Sciences.

5. **Adjunct Lecturer**, Department of Computer Science, FAST National University of Computer and Emerging Sciences, Islamabad, Pakistan 2015–2016

Undergraduate level courses: Business Statistics; Probability and Statistics.

Research Grants and Funding Awards

1. **2021:** Jan Wallanders och Tom Hedelius stiftelse, Tore Browaldhs stiftelse. 208,000 SEK. Personal travel grant to visit University of Alabama, Tuscaloosa, USA. (Ref: H2021-0027)
2. **2024:** Research grant from Bertil och Britt Svenssons Stiftelse för Belysningsteknik, 325,000 SEK, together with Professor Myriam Aries, Department of Construction Engineering and Lighting Science, JTH, Jönköping University.

Published Journal Articles

1. Iqbal, S., Hussain, Z., **Omer, T.** (2026). *Some New Quantitative Randomized Response Models using Optional and Partial Scrambling for Sensitive Data.* *Scientific Reports.*
2. **Omer, T.**, Månsson, K., Sjölander, P., & Salah Uddin, G. (2026). *Machine Learning Approaches to Forecast the Realized Volatility of Crude Oil Prices.* *Journal of Forecasting* (Wiley).
3. **Omer, T.**, Sjölander, P., Månsson, K., & Kibria, B. M. G. (2024). *Improved Breitung and Roling estimator for mixed-frequency models with application to forecasting inflation rates.* *Statistical Papers* (2024): 1–23.
4. **Omer, T.**, Sjölander, P., Månsson, K., & Kibria, B. M. G. (2021). *Improved Estimators for the Zero-Inflated Poisson Regression Model in the Presence of Multicollinearity: Simulation and Application of Maternal Death Data.* *Communications in Statistics: Case Studies, Data Analysis and Applications*, 7(3), 394–412. DOI: 10.1080/23737484.2021.1952493.
5. **Omer, T.**, Hassan, M. U., Hussain, I., Ilyas, M., Hashmi, S. G. M. D., & Khan, Y. A. (2022). *Optimization of monitoring network to the rainfall distribution by using stochastic search algorithms: lesson from Pakistan.* *Tellus A: Dynamic Meteorology and Oceanography*, 74(1).
6. Qasim, M., Amin, M., & **Omer, T.** (2020). *Performance of some new Liu parameters for the linear regression model.* *Communications in Statistics—Theory and Methods*, 49(17), 4178–4196.
7. **Omer, T.**, Zawar, H., Qasim, M., Khan, A. A., & Shah, F. S. (2021). *Two Different Classes of Shrinkage Estimators for the Scale Parameter of the Rayleigh Distribution.* *Journal of Modern Applied Statistical Methods*, 19(1). DOI: 10.22237/jmasm/1608553440.

8. Khan, M., **Omer, T.**, Ellahi, A., Ur Rahman, Z., Niaz, R., & Ahmad Lone, S. (2023). *Monitoring and assessment of heavy metal contamination in surface water of selected rivers*. Geocarto International, 1–17.
9. Raza, M. A., Almazah, M. M., Hussain, I., Al-Duais, F. S., Al-Rezami, A. Y., & **Omer, T.** (2023). *Bayesian logistic regression analysis for spatial patterns of inter-seasonal drought persistence*. Geocarto International, 38(1), 2211041.
10. Niaz, R., Almazah, M. M., Al-Rezami, A. Y., Ali, Z., Hussain, I., & **Omer, T.** (2023). *Proposing a new framework for analyzing the severity of meteorological drought*. Geocarto International, 38(1), 2197512.
11. Noreen, K., **Omer, T.**, ul Hassan, J., Rasheed, H. K., & Ahmed, R. (2023). *Some New Constructions of Minimal Efficient Circular Nearly Strongly Balanced Neighbor Designs*. Journal of King Saud University—Science, 102748.
12. Shabbir, W., **Omer, T.**, & Pilz, J. (2023). *The impact of environmental change on landslides, fatal landslides, and their triggers in Pakistan (2003–2019)*. Environmental Science and Pollution Research, 30(12), 33819–33832.
13. Noreen, K., Rashid, M. S., Shehzad, F., Ul Hassan, M., Noreen, Z., **Omer, T.**, & Ahmed, R. (2022). *Algorithms to obtain generalized neighbor designs in minimal circular blocks*. Communications in Statistics—Simulation and Computation, 1–12.
14. Qasim, M., Amin, M., Akram, M. N., **Omer, T.**, & Hussain, F. (2019). *Forecasting Buffalo Population of Pakistan using Autoregressive Integrated Moving Average (ARIMA) Time Series Models*. Proceedings of the Pakistan Academy of Sciences: A. Physical and Computational Sciences, 56(3), 27–36.
15. Syed, A., Zhang, J., Moniruzzaman, M., Rousta, I., **Omer, T.**, & Olafsson, H. (2021). *Situation of urban mobility in Pakistan: before, during, and after the Covid-19 lockdown with climatic risk perceptions*. Atmosphere, 12(9).

Manuscripts In reviews

1. **Omer, T.** (2025). *MIDAS Poisson regression advancements in Dengue fever prediction using Google Trends and environmental data*. One health(In review).
2. Khan, M. I., **Omer, T.**, Månsson, K., & Sjölander, P. (2025). *Forecasting Financial Volatility: A Dual-Parameter Heterogeneous Autoregressive Model for Realized Volatility*. Statistical Papers (In review).
3. **Omer, T.**, Henderson, D. J., & Kourtellos, A. (2025). *Multidimensional Panel Data Regression Model: The Case of the Multidimensional Home Ownership Vacancy Rate in the USA*. Journal of Business & Economic Statistics (In review).
4. **Omer, T.**, Henderson, D., Qasim, M., & Månsson, K. (2025). *Modeling Asymmetric Noise in Regulatory Benchmarking: Evidence from Swedish Electricity Distribution*. Energy Policy (In review).

Working Papers and Ongoing Projects

1. **Omer, T.**, Henderson, D. J., & Kourtellos, A. *An RU MIDAS Trajectories-Based Approach Measuring Intergenerational Mobility in Sweden* Working on initial draft (2025).
2. Khan, M. I., **Omer, T.**, & Salah Uddin, G. (2026). *Forecasting the Vector Heterogeneous Autoregressive (VHAR) Models with Machine Learning Techniques Using the Sentiment Index*. (Working paper)
3. **Omer, T.**, Javed, F., & Qasim, M. (2026). *Lasso and Adaptive Lasso Methods in Sparse Mixed Data Sampling Regressions*. (Working paper)

Conference Presentations and Proceedings

- **Omer, T.**, Månsson, K., Sjölander, P., & Kibria, B. M. G. (2021). *An Unrestricted MIDAS Poisson Regression Model*. CFE–CMStatistics 2021, King’s College London, UK, December 18–20, 2021.
- **Omer, T.**, Månsson, K., Sjölander, P., & Kibria, B. M. G. (2021). *Improved Boosted Two-Parameter Breitung Estimator: An Application to the Crude Oil Prices*. JSM Proceedings, Business and Economic Section, American Statistical Association, 1716–1726.
- **Omer, T.**, Sjölander, P., Månsson, K., & Kibria, B. M. G. (2019). *Improved Estimators for the Zero-Inflated Poisson Regression Model in the Presence of Multicollinearity*. CM-Statistics 2019, University of London, December 14–16, 2019.
- Qasim, M., Amin, M., Azam, M., & **Omer, T.** (2019). *On Almost Unbiased Ridge Estimator in the Poisson Regression Model*. 2nd Asia-Pacific Conference on Applied Mathematics and Statistics, University of Malaya, Kuala Lumpur, Malaysia, February 21–24, 2019.
- **Omer, T.**, Hussain, Z., & Qasim, M. (2019). *Optimized Monitoring Network of Pakistan by using stochastic search algorithms*. 11th RSEP International Multidisciplinary Conference, Vienna, Austria, April 9–11, 2019.
- **CFE–CMStatistics 2025 (Hybrid)**, Birkbeck, University of London, UK, December 13–15, 2025. *Presentation Certificate: CC1378, Multidimensional panel data regression model: The case of the multidimensional homeownership vacancy rate in the USA*. Presenting: Talha Omer. Co-authors: Daniel Henderson, Andros Kourtellos.

Participation in Workshops and Tutorials

- July 17–21, 2023: Five-day course on *Microeconomic Forecasting and Analysis in the Machine Learning Era*, Braga, Italy.
- June 27–30, 2022: Four-day summer course on *The Econometrics of Big Data*, delivered by Prof. Christian B. Hansen, University of Minho, Braga, Portugal.
- December 6–8, 2019: Workshop at the Center for Behavioral Experiments (CEBEX) Winter School on *Data and Behavioral Economics for Data Analysis in R*, Prague, Czech Republic.
- December 13, 2019: Tutorials on R programming and mixture models with applications to image analysis at Senate House, University of London, UK. Taught by Prof. Jochen Einbeck, Durham University, UK.
- November 8, 2019: Workshop on *JIBS Teacher Certificate Seminar*, Jönköping University, Sweden.
- April 25–26, 2018: Organized a one-day workshop on *Endnote x7 Basic*.
- January 22–24, 2018: Three-day training on *Andragogical/Pedagogical Skills*, PHEC Staff Development Academy, Lahore.

Research Visits

- **2024:** Two-week research visit to the Department of Economics, University of Cyprus, to collaborate with Professor Andros Kourtellos.
- **2022–2023:** Approximately six-month research visit to work with Professor Daniel Henderson at the University of Alabama, Tuscaloosa, USA.

Pedagogical Competence (Swedish Higher Education)

Completed **7.5 ECTS** in Academic Teaching and Learning at Jönköping University, consisting of five courses (each 1.5 ECTS): Self-contained Student Active Learning; Conditions and Pedagogical Perspectives; Online Education; Examination and Assessment; Broadened Participation.

Invited Talks

- “Introduction to machine learning algorithms” in “Introduction to Statistical Learning and Machine Learning for Economics and Finance”, Department of Management, Linköping University, Sweden (Onsite). December 03, 2025.
- “Introduction to machine learning algorithms” in “Introduction to Statistical Learning and Machine Learning for Economics and Finance”, Department of Management, Linköping University, Sweden (Onsite). December 09, 2024.
- “Introduction to machine learning algorithms and high dimensional statistics methods”, Department of Economics and Quantitative Methods, University of Management and Technology, Lahore, Pakistan (Online). December 10, 2024.

Data Analyst Experience

- **2014–2015: Research Assistant**, Under the HEC National Research Project University (NRPU) titled “Optimization of Meteorological Monitoring Network in Pakistan”, Funded by HEC Pakistan.
- **June 2014–September 2014: Data Analyst**, BTK Associates Rawal Dam Colony, Islamabad, BRSP Project.

PROFESSIONAL AFFILIATIONS AND REVIEW ACTIVITIES

- Communication in Statistics-Theory and Methods
- Environmental Science and Pollution Research
- Communication in Statistics-Simulation and Computations
- Economic Modelling
- PLOS One
- Journal of King Saud University of Science

Computer Skills

- **Statistical Software & Language:** R, Python, STATA, Tableau, SPSS, Minitab, Statistix 8.1
- **Computer Applications:** MS Word, MS Excel, MS PowerPoint, LaTeX

Extracurricular Interests

Playing badminton and cricket (professional).

Online Profiles

- **Google Scholar:** Google Scholar
- **ResearchGate:** ResearchGate
- **Personal Website:** Personal Website

References

- **Pär Henrik Sjölander:** Professor of Econometrics, Department of Economics, Finance and Statistics, JIBS, Jönköping University, Sweden.
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- **Kristofer Månsson:** Professor of Statistics, Department of Economics and Statistics, Linnaeus University, Sweden.
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- **Gazi Salah Uddin:** Professor of Economics, Department of Management and Engineering (IEI), Linköping University, Sweden.
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